LEARNER GUIDE

Converting spatial units

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

When you change units, length scales by 10, but area scales by 100 and volume by 1000. In this lesson you zoom into a 1 cm length, a 1 cm^2 square and a 1 cm^3 cube to reveal the millimetre units hiding inside — 10 in the line, 100 in the square, 1000 in the cube. You fill a square and a cube unit by unit to feel where those numbers come from, and you convert real measurements between mm, cm and m. The question you are exploring is: why does 1 cm^2 become 100 mm^2 — not 10?

What you are learning to notice

- That 1 cm = 10 mm, 1 cm^2 = 100 mm^2, and 1 cm^3 = 1000 mm^3.
- That each extra dimension multiplies the scale factor by another 10 — powers of ten (10, 10^2, 10^3).
- How to convert lengths, areas and volumes between mm, cm and m, both ways.
- That the same idea reaches metres: 1 m^2 = 10 000 cm^2 and 1 m^3 = 1 000 000 cm^3.

Moving through it, one step at a time

In Zoom & Reveal, choose length, area or volume and press Zoom in to see the millimetre units inside a centimetre unit — and count them. In Fill It Up, pack a square with mm^2 and a cube with mm^3, a row or layer at a time, until you reach 100 and 1000. In Spot the Slip, decide whether each conversion is right. Then convert real measurements — a floor, a screen, a tank. There is no rush; use the grid on the Scratchpad to draw a cm square as 100 mm squares.

A gentle reminder

You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (multiplying and dividing by 10, 100 and 1000, and the idea of powers of ten), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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